Other revenue increased by $24.3 million, or 40%, during fiscal year 2019 as compared to fiscal year 2018, primarily due to the increase in product sales and professional services resulting from the overall growth in our business.

Cost of Revenues and Gross Margin

	Year ended December 31,				Year ended December 31,			
(in thousands, except percentages)	2019	2018	$ Change	% Change	2018	2017	$ Change	% Change
Cost of revenues								
Subscriptions	$ 160,320	$ 109,454	$ 50,866	46%	$ 109,454	$ 89,193	$ 20,261	23%
Other	70,723	47,675	23,048	48%	47,675	32,078	15,597	49%
Total cost of revenues	$ 231,043	$ 157,129	$ 73,914	47%	$ 157,129	$ 121,271	$ 35,858	30%
Percentage of revenues								
Subscriptions	18%	16%			16%	18%		
Other	8%	7%			7%	6%		
Gross margins								
Subscriptions	80%	82%			82%	81%		
Other	17%	22%			22%	16%		
Total gross margin %	74%	77%			77%	76%		

Subscription cost of revenues and gross margin. Cost of subscriptions revenues increased by $50.9 million, or 46%, during fiscal year 2019 as compared to fiscal year 2018. Primary drivers of the increase were increases in third-party costs to support our solution offerings of $21.5 million, infrastructure support costs of $19.8 million including amortization expense from acquired intangible assets, and headcount and personnel and contractor related costs of $9.6 million including share-based compensation expense. These factors resulted in a decrease in gross margin.

The increase in headcount and other expense categories described herein was driven primarily by investments in our infrastructure and capacity to improve the availability of our subscription offerings, while also supporting the growth in new customers and increased usage of our subscriptions by our existing customer base. We expect subscription gross margin to be within a relatively similar range in the future.

Other cost of revenues and gross margin. Cost of other revenues increased by $23.0 million, or 48%, during fiscal year 2019 as compared to fiscal year 2018. This was primarily due to the increase in services personnel costs of $11.1 million including share-based compensation expense, cost of product sales of $10.6 million, and overhead costs of $1.3 million. Other revenues gross margin fluctuates based on timing of completion of professional services projects and discounting on phones.

Research and Development

	Year ended December 31,				Year ended December 31,			
(in thousands, except percentages)	2019	2018	$ Change	% Change	2018	2017	$ Change	% Change
Research and development	$ 136,363	$ 101,042	$ 35,321	35%	$ 101,042	$ 75,148	$ 25,894	34%
Percentage of total revenues	15%	15%			15%	15%		

Research and development expenses increased by $35.3 million, or 35%, during fiscal year 2019 as compared to fiscal year 2018, primarily due to increases in personnel and contractor costs of $30.3 million and overhead costs to support our research and development efforts of $4.8 million. Of the total increase in personnel and contractor costs, approximately $20.0 million was primarily driven by headcount growth and $8.2 million was due to higher share-based compensation expense.

The increases in research and development headcount and other expense categories were driven by continued investment in current and future software development projects for our applications. Given the continued emphasis and focus on product innovation, we expect research and development expenses to continue to increase in absolute dollars.